

- Aircraft structures
1. Aerostructure design
  2. Structural examples
  3. Fasteners and bonding

# 2B: Aircraft structures supplement

Lecture Presented By: Kevin Kochersberger

In this lecture you will learn:

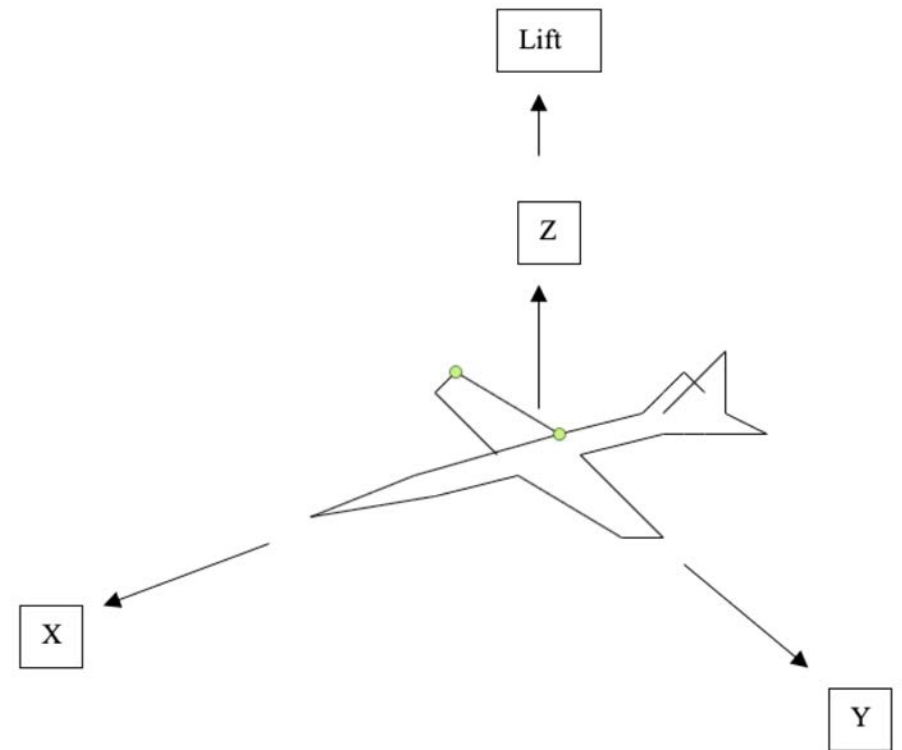
- How aircraft structures are designed
- Examples of typical aircraft structures

# The design of an aircraft structure starts by considering the loads that must be carried

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- A load factor in each orthogonal direction is determined from the flight maneuvers
  - The aircraft is designed to carry these loads without failure
- Load factors in the  $x$ ,  $y$ , and  $z$  directions are expressed as  $n_x$ ,  $n_y$  and  $n_z$ 
  - The load factor is a multiplier to the weight ( $W$ ) of the aircraft

$$\sum F_x = n_x W \quad \sum F_y = n_y W \quad L = n_z W$$



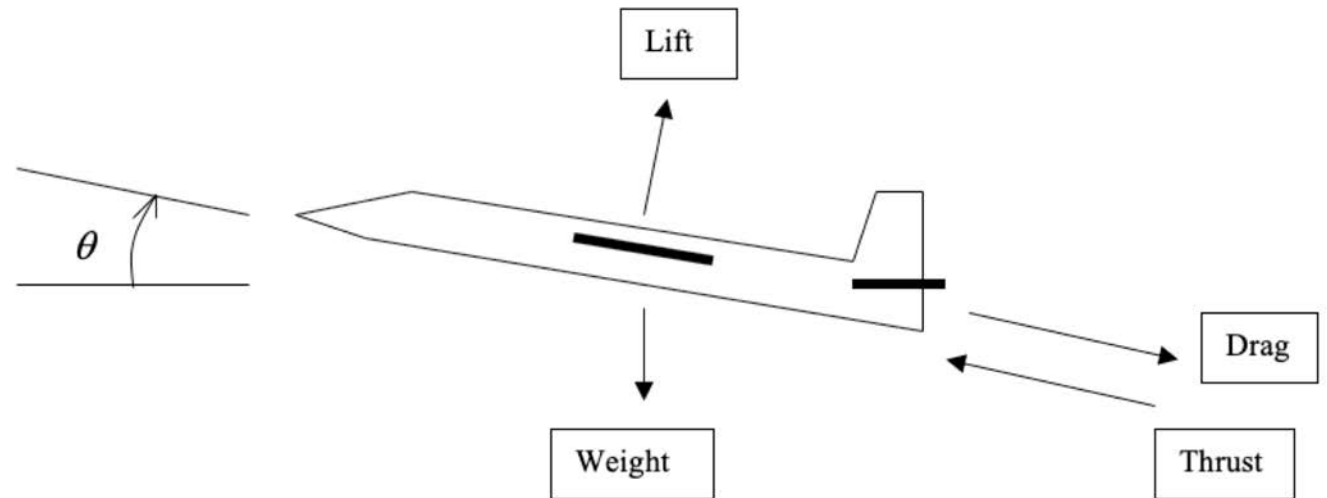
# Loading above non-accelerated, level flight is caused by flight loads through maneuvering and gusts

## Aircraft structures

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- As the aircraft maneuvers and responds to gusts, it is subject to additional loading

$$n_z = \frac{a_z}{g} + \cos \theta$$
$$n_x = \frac{\sum F_x}{W} = \frac{a_x}{g}$$
$$n_y = \frac{\sum F_y}{W} = \frac{a_y}{g}$$

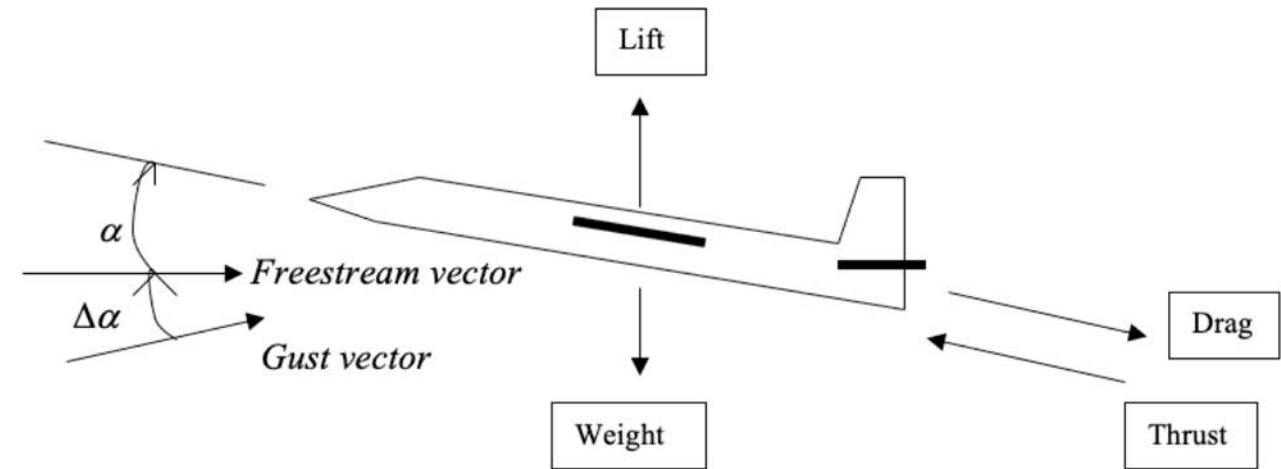


# Vertical gusts cause additional loading which can be defined by considering the vertical component of wind

## Aircraft structures

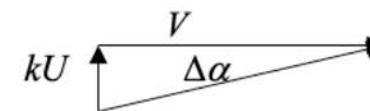
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- As the aircraft maneuvers and responds to gusts, it is subject to additional loading
- A vertical gust of magnitude  $kU$  causes an additional angle of attack on the wing



The change in angle of attack is related to the *vertical gust velocity*  $U$ , the *shape of the gust*, and the *freestream velocity*  $V$ :

$$n_z = 1 + \frac{\rho V C_{L\alpha} k U}{2W/S}$$



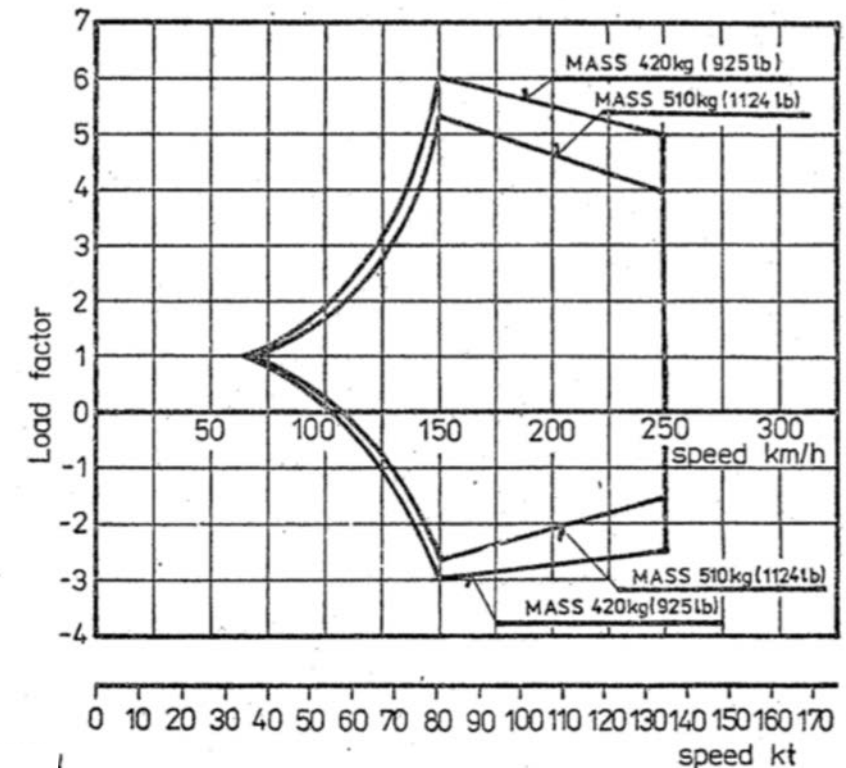
# The relationship between load factor if the aircraft encounters a gust, and aircraft speed, is expressed in a $V-n$ diagram

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- The  $V-n$  diagram shows the *maximum vertical load factor* that would be encountered if the airplane generated the maximum lift that it was capable of generating (at  $C_L$  max)
- Note that as the airspeed increases, the potential to generate large loads increases
- The speed of 150 km/h marks the point at which maneuvering must be limited to keep the load factor below 6

2.7. MANOEUVRING LOAD FACTORS

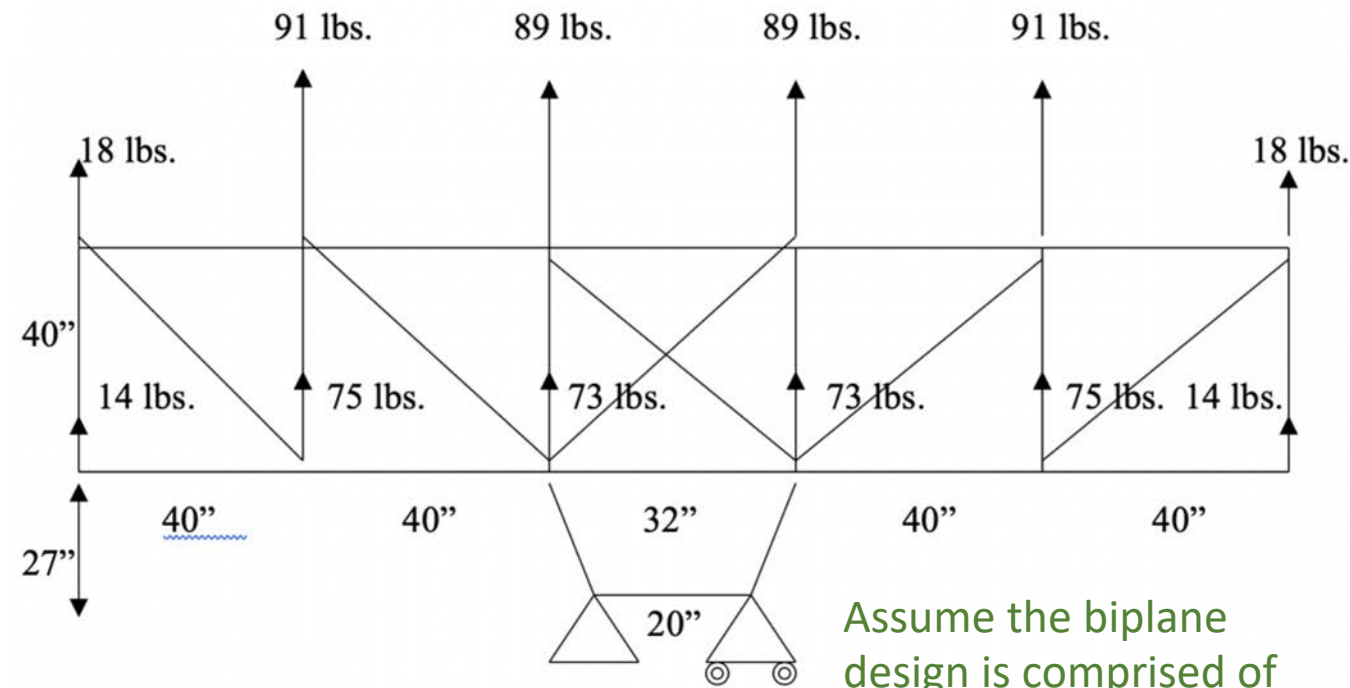


# The load factor directly tells us what loads will be applied to the aircraft

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- In the aircraft structural design, it is important to **design the structure to support the design loads** and **include a factor of safety** to ensure the aircraft does not fail
- The **factor of safety (FS)** is typically **1.5 – 2.0** for aircraft
- Simplifying analyses can be performed to ensure the aircraft structure is adequate



Assume the biplane design is comprised of pinned joints with two-force members only

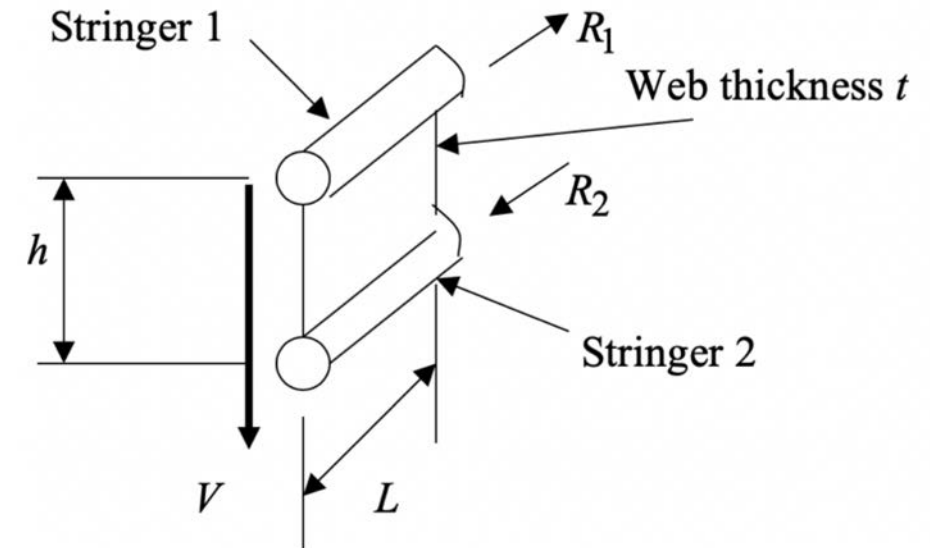
# The wing spar carries most of the wing loads

- Many structures in aircraft can be analyzed as beams in bending
- The **wing spar** is comprised of **flanges or stringers** on the top and bottom, and a **web** in the middle
- The loads generated by the wing in lift are transferred to this spar



## Aircraft structures

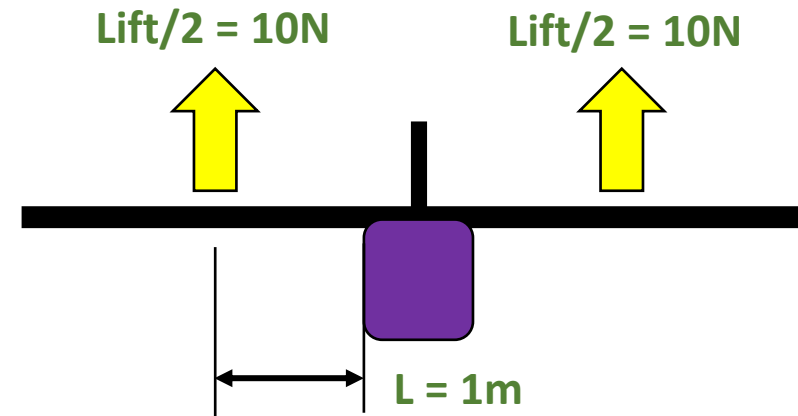
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A stringer-web structure is commonly found in wings and is a simplified model for analyzing wing spars

# The lift can be approximated as a resultant vector in the center of the wing

- For the wing spar section shown on the right, if the wing has a 4m wingspan, then we can approximate the max stress in the top and bottom flanges
- In **straight and level flight**, if the aircraft weighs 20 N, then the lift generated by the wings is 20 N
- If the aircraft experiences a maneuver that results in a **vertical acceleration of 3g**, the lift generated by the wings is:



**Straight and level flight:**

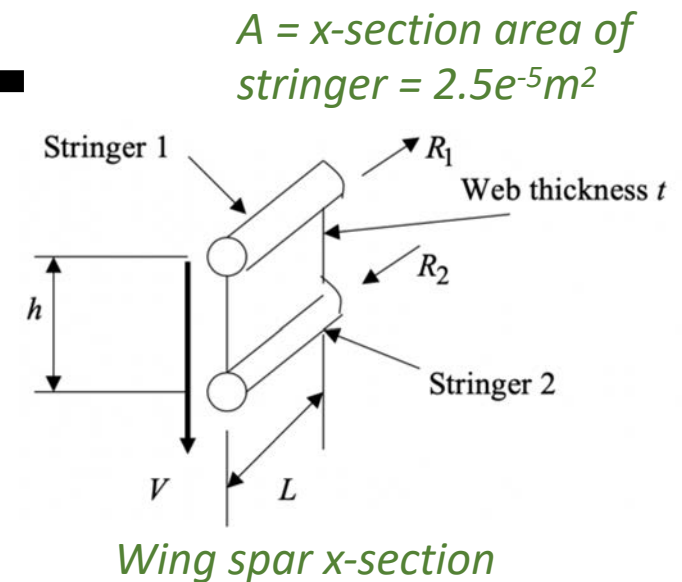
$$Lift = W = 20N$$

**3g maneuver:**

$$Lift = Wn_z = 20N * 3 = 60N$$

Aircraft structures

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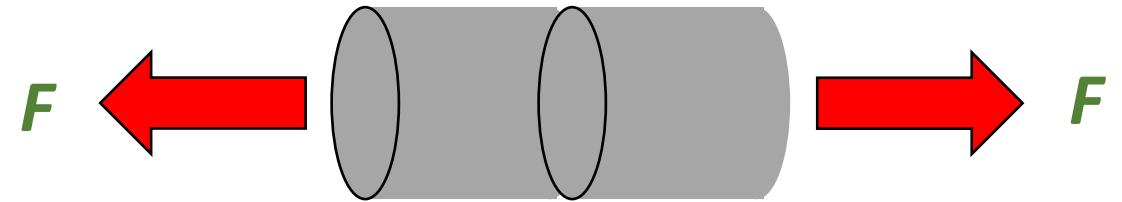


# Stress is the force/area that the spar material experiences

## Aircraft structures

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- A factor of safety is applied to account for factors that are not possible to model
  - Manufacturing imperfections
  - Material defects
  - Assembly errors – may cause higher-than-expected loads on the airframe
- If (computed stress)( $FS$ )  $< S_y$  the material will not fail



$A$

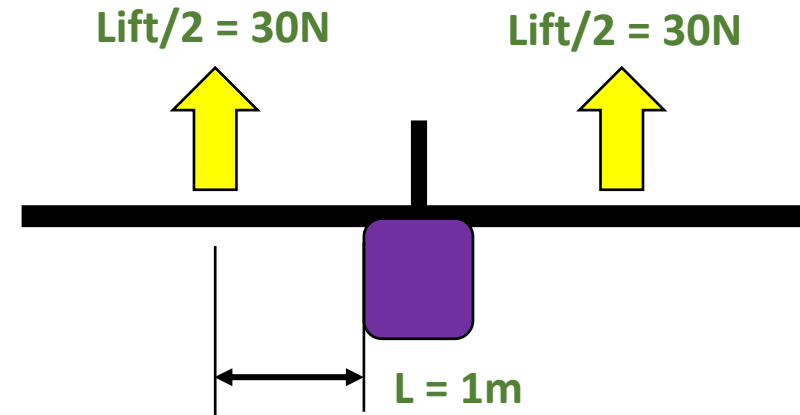
$$\sigma = \frac{F}{A}$$

$$FS \leq \frac{S_y}{\sigma}$$

$S_y$  = Material yield strength in MPa

# Stress is the force/area that the spar material experiences

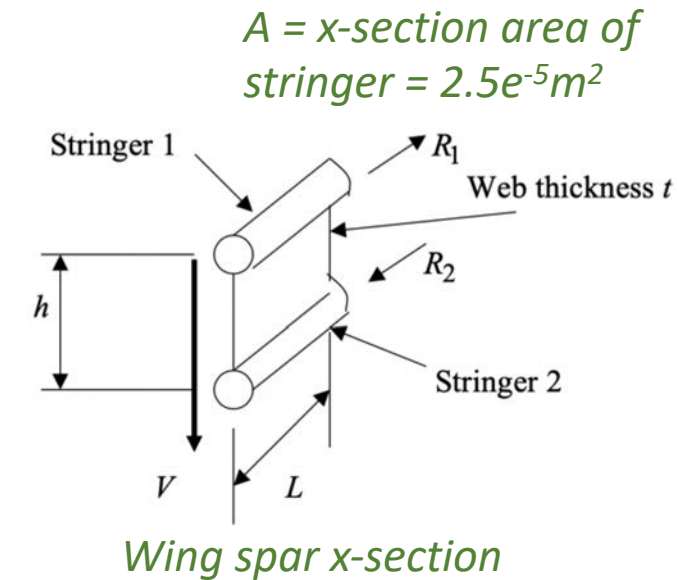
- For the wing spar section shown on the right, if the wing has a 4m wingspan, then we can approximate the **max stress in the top and bottom flanges** as:



$$\sigma = \frac{\text{Moment} * h}{2I} = \frac{Mh}{2I} = \frac{\frac{\text{Lift}}{2} * L * h}{2(\frac{Ah^2}{2})} = \frac{\text{Lift} * L}{2Ah} = \frac{60Nm}{2 * 2.5e^{-5} * .02}$$
$$\sigma = 60 MPa$$

## Aircraft structures

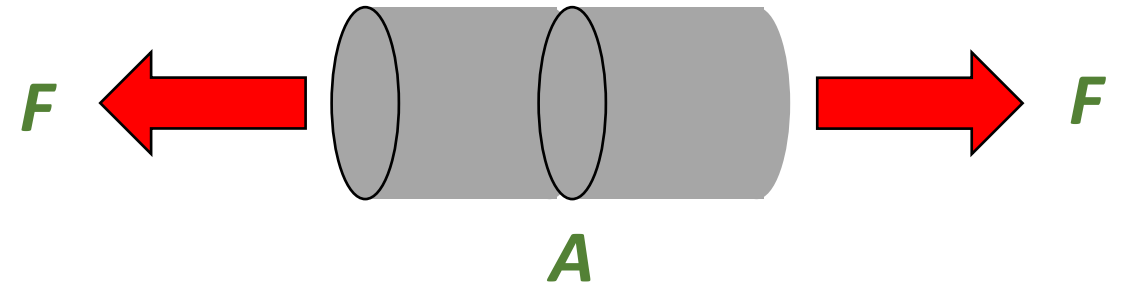
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# The resulting stress is compared to the material strength to determine if the design is safe

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- 6061 Aluminum, a popular aluminum used in aircraft structures, has a yield strength  $S_y = 290 \text{ MPa}$
- If the required factor of safety is  $FS = 2$ , then we can check to see if the aircraft is designed with adequate strength



$$FS \leq \frac{S_y}{\sigma}$$

$$2.0 \leq \frac{290}{60} = 4.8$$

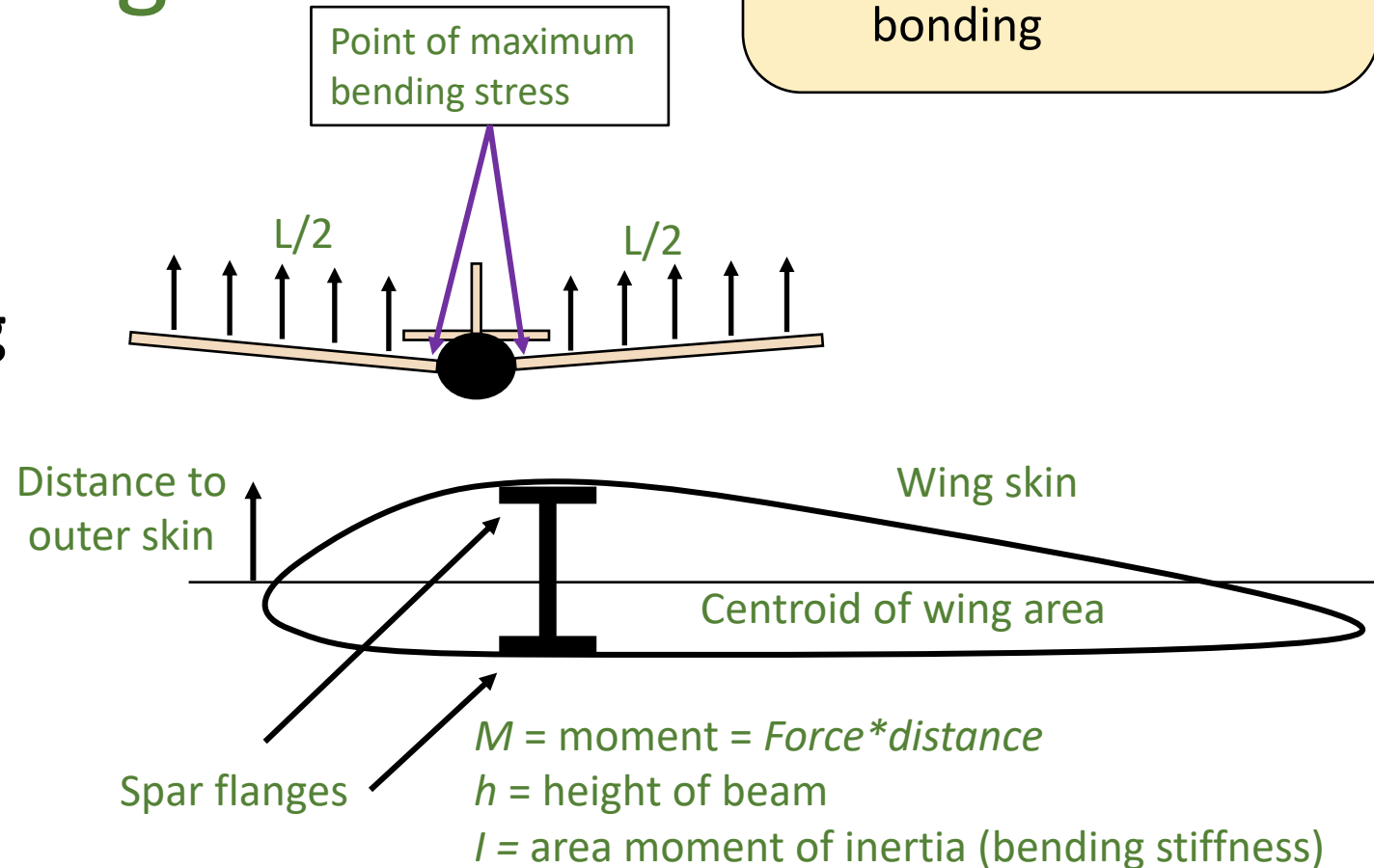
*Because the computed factor of safety (4.8) is greater than the required factor of safety (2.0), the design is good!*

# Aircraft designs must maximize strength and reduce weight

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- This is accomplished in a structural design that **maximizes the distance of wing material** from the centroid to the **outer structure**
- Wings are made **strong and lightweight** by designing the wing skin to carry loads like the spar flanges

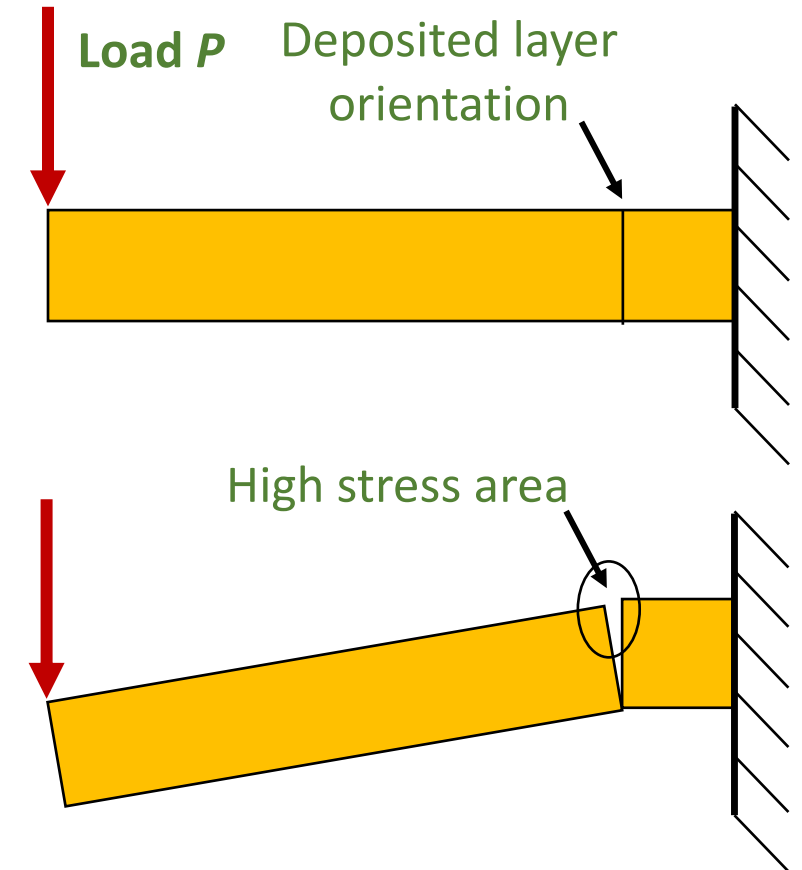


$$\sigma = \frac{Mh}{2I}$$

# 3D printed structures are not always printed in the optimal orientation which results in a reduction of strength

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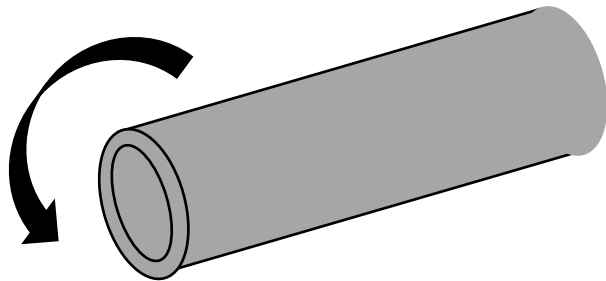
- Care must be taken to print parts so that high stress areas are not unfavorably oriented
- **High loads** that are **perpendicular to the print direction** should be avoided as this is a weak orientation for 3D printed parts
- **Bending loads result in high normal tension loads** at the extreme sides of structures where failure is more likely to occur
- Prints should consider the normal stress due to bending



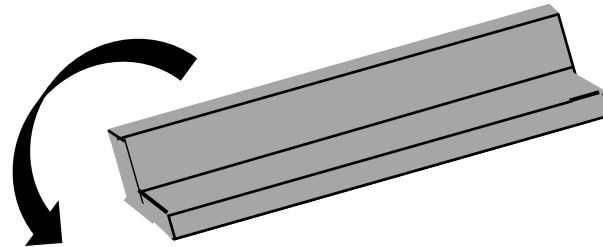
# Simple rules can be used in creating strong but lightweight structures

- Torsional loads are best carried by “closed section” structures
  - These are structures that have a cross section which is continuous
  - Open sections (such as L or C shapes) cannot carry torsional loads effectively

Applied torque  $T$



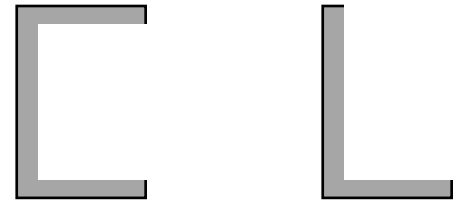
Good!



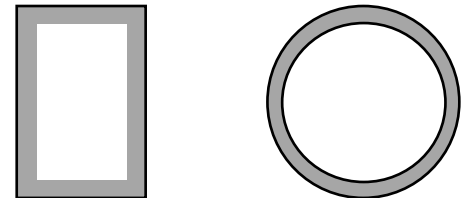
Bad!

## Aircraft structures

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Cross sections that are “open”



Cross sections that are “closed”

# Simple rules can be used in creating strong but lightweight structures

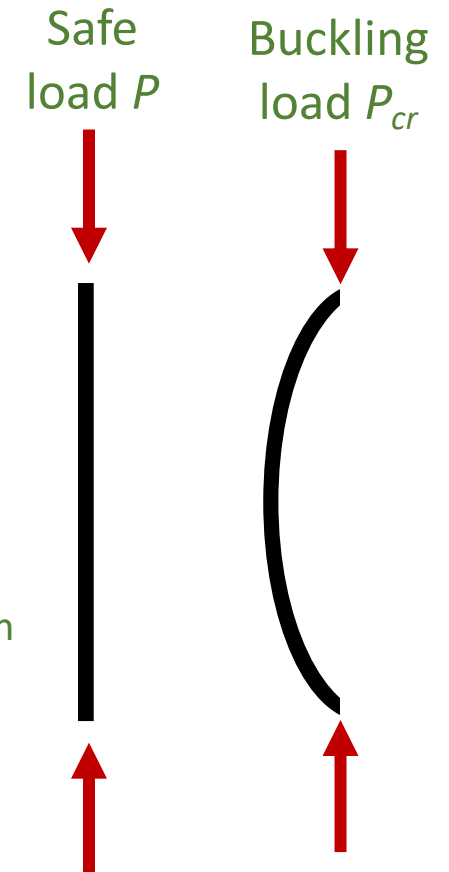
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- Thin columns that carry axial loads are **effective in tension** but **not effective in compression**
- Compression loads in thin columns can cause buckling to occur
  - Buckling is a condition of instability of the member and results in a catastrophic failure of the structural member
- Reducing the column length or load will reduce the likelihood for failure

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

$E$  = Modulus of elasticity (stiffness of material)  
 $I$  = Cross section area moment of inertia  
 $K$  = Column effective length factor (=1 for the pinned example shown)  
 $L$  = Column length



# Buckling example

- PLA (Polylactic Acid) is a common 3D printing thermoplastic material
- The datasheet for PLA indicates  $E = 2.3 \text{ GPa}$
- If a column is 5 cm long and 4 mm in diameter, determine the critical buckling load assuming pinned connections

$$I = \frac{\pi D^4}{64} = 1.26e^{-11}$$

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} = \frac{\pi^2 (2.3e^9) 1.26e^{-11}}{(0.05)^2} = 114 \text{ N}$$

$P$



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PLA  
(Polylactic Acid)



Technical Data Sheet

| Tensile Properties<br>ASTM D638 - Type V                                                                                                                                                                |                                            |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------|
| Property                                                                                                                                                                                                | Imperial                                   | Metric                      |
| Toughness*                                                                                                                                                                                              | 7.7 ft-lb/in <sup>2</sup>                  | 16.2 KJ/m <sup>2</sup>      |
| Tensile Modulus                                                                                                                                                                                         | 293000 psi                                 | 2.3 GPa                     |
| Ultimate Tensile Strength                                                                                                                                                                               | 7080 psi                                   | 26.4 MPa                    |
| Tensile Strength at Yield                                                                                                                                                                               | 8840 psi                                   | 35.9 MPa                    |
| Elongation at Yield                                                                                                                                                                                     | 2%                                         | 2%                          |
| Elongation at Break                                                                                                                                                                                     | 4%                                         | 4%                          |
| 3D Printing Properties                                                                                                                                                                                  |                                            |                             |
| Property                                                                                                                                                                                                | Imperial                                   | Metric                      |
| Expected Max Linear Print Speed                                                                                                                                                                         | 3.54 in/s                                  | 90 mm/s                     |
| Hardness, ASTM D2240                                                                                                                                                                                    | 95D                                        | 95D                         |
| Solid Density, ASTM D792                                                                                                                                                                                | 4.48 x 10 <sup>-2</sup> lb/in <sup>3</sup> | 1.24 g/cc                   |
| Impact Properties                                                                                                                                                                                       |                                            |                             |
| Property                                                                                                                                                                                                | Imperial                                   | Metric                      |
| Notched Izod (machined), 23 C, ASTM D256                                                                                                                                                                | 0.3 ft-lb/in                               | 16 J/m                      |
| Gardner Impact, 23 C, ASTM D5420                                                                                                                                                                        | 10.3 ft-lb                                 | 14 J                        |
| Thermal Properties                                                                                                                                                                                      |                                            |                             |
| Property                                                                                                                                                                                                | Imperial                                   | Metric                      |
| Glass Transition by DSC, ASTM E1356                                                                                                                                                                     | 134 F                                      | 57 C                        |
| Glass Transition by DMA, ASTM D792                                                                                                                                                                      | 145 F                                      | 63 C                        |
| Heat Deflection Temperature, ASTM D648                                                                                                                                                                  | 121 F                                      | 49 C                        |
| Coefficient of Thermal Expansion, ASTM E832                                                                                                                                                             | 23 x 10 <sup>-6</sup> in/inR               | 41 x 10 <sup>-6</sup> m/m-K |
| Heat Capacity, ASTM E1269                                                                                                                                                                               | 0.43 Btu/lb/°F                             | 1,800 J/kg-K                |
| Thermal Conductivity, ASTM C518                                                                                                                                                                         | 0.9 Btu-in/hr/ft <sup>2</sup> /°F          | 0.13 W/m-K                  |
| Available Colors                                                                                                                                                                                        |                                            |                             |
| Black, Blue, Green, Grey, Natural, Orange, Red, White, Yellow                                                                                                                                           |                                            |                             |
| Suggested Uses                                                                                                                                                                                          |                                            |                             |
| PLA is one of the most cost-effective FDM printing materials and easiest to process. PLA is best used for complex modeling applications that do not require impact strength or tolerance to heat loads. |                                            |                             |

\*Toughness is not defined in ASTM D638 though can be calculated by taking the integral of the stress-strain curve collected by tensile data.



# Buckling example

- Note that the load carrying capacity is higher when the column is analyzed in tension

$$\sigma = \frac{P}{A} = \frac{P}{\pi r^2} = \frac{P}{\pi (.002^2)} = 79580 P$$

$$S_y = 26.4 \text{ MPa}$$

- For a factor of safety of 2:

$$2(\sigma) \leq S_y$$

$$\sigma \leq 13.2 \text{ MPa}$$

$$P \leq 166 \text{ N}$$

- The acceptable load in tension is 166 N vs. 114 N when loaded in compression

$P$



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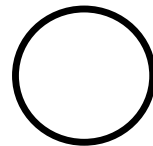
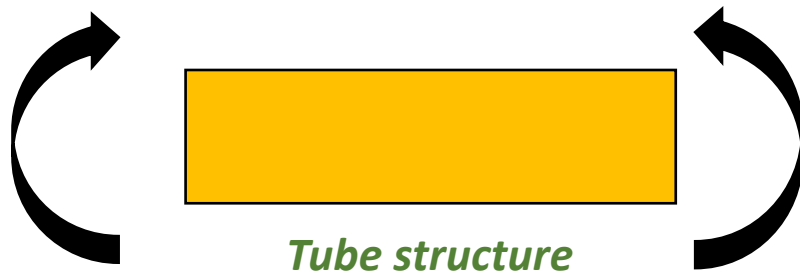
# Monocoque design refers to the use of the outer skin of a structure to carry the design loads

- **Structure - monocoque design**

- The loads are carried through the outer shell of the aircraft
- This makes a very lightweight, strong structure
- Aerodynamically it works well for fixed wing aircraft
  - The DJI Phantom series aircraft are made in this fashion

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*Loads are carried in the outer shell efficiently*

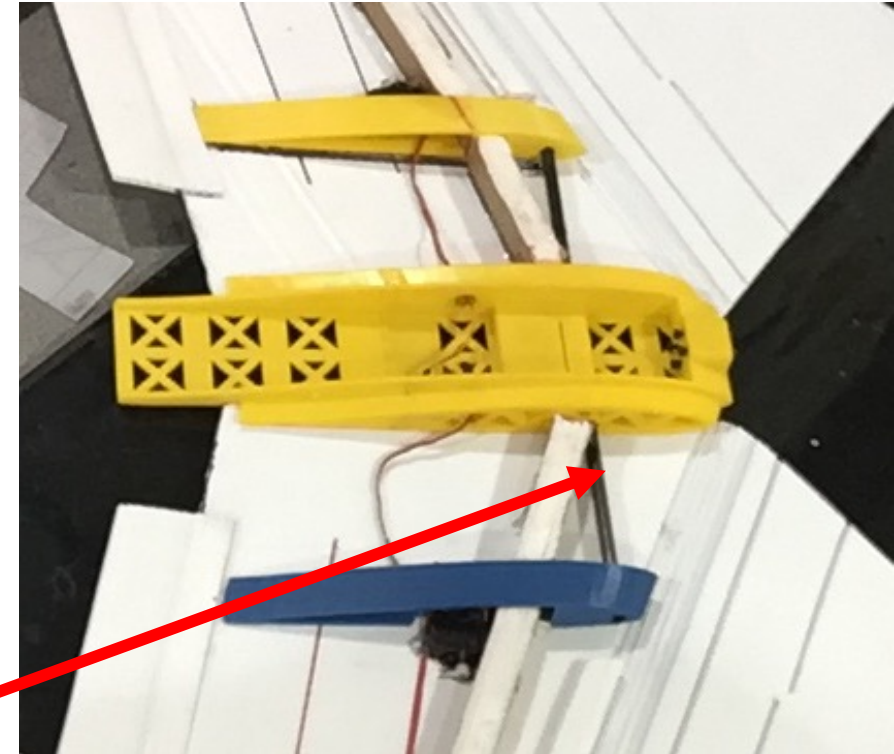


# The construction of an airframe should keep the concept of *load path* in mind

- Load paths refer to the way in which loads are carried throughout an aircraft structure
- It is important to design a structure such that load paths are efficiently guided around the airframe through the main load-carrying members
  - Sometimes a design has failed to take into consideration how loads are transmitted and this can cause an unexpected failure

*An early-generation EcoSoar shows the carbon spar that passes through the middle of the fuselage body, carrying loads between the wings and not loading the 3D printed fuselage directly*

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  3. **Fasteners and bonding**



# Wing construction methods

- Carbon epoxy is a common aircraft construction material because it is strong and lightweight, and it can be molded into a desired shape
- The methods of carbon epoxy construction include wet layup and pre-preg molded construction
- Carbon is typically in the form of a cloth, and epoxy is a two-part system that cures to a hard finish
  - The combination of carbon and epoxy results in a very strong finished product

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# Wing construction methods

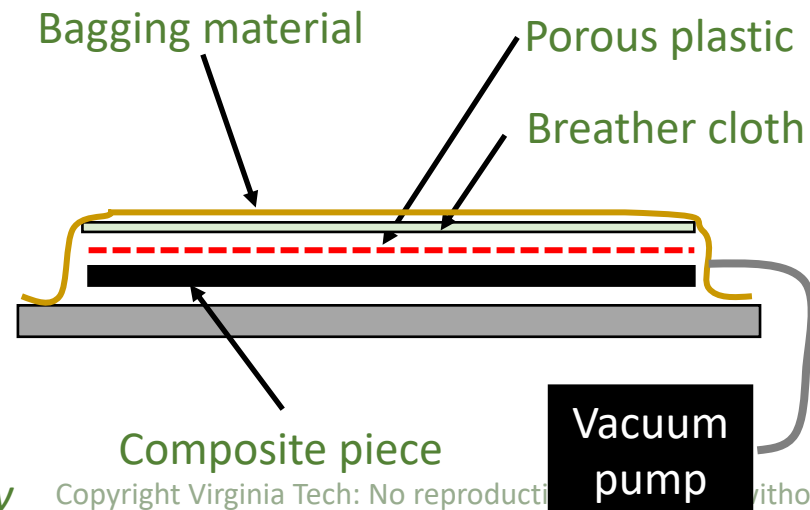
- **Wet layup** is a technique of laying down carbon cloth in the desired shape and applying epoxy so the system cures in the desired shape
- It is relatively simple to create complex shapes
- Forms can be used to improve the quality of the epoxy construction, and vacuum bagging is typically used to apply pressure and remove excess epoxy

## Aircraft structures

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*Two carbon tubes joined in a T-joint by laying up carbon cloth and applying epoxy*



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*Vacuum bag to apply pressure to carbon on mold*



# Wing construction methods

- A method by which the part shape is first produced by machining wax (easy to machine) is commonly used:
  - The part shape is produced in an easy-to-machine material
  - An external mold is created by laying up fiberglass epoxy on this part
  - The fiberglass mold is then used as a form to lay up the final carbon epoxy product

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*The wax plugs on the left are machined using a CNC mill. Fiberglass-epoxy is vacuum-bagged to the plug to create a fiberglass mold (right). Carbon epoxy is laid up in the fiberglass mold and vacuum bagged. The finished product is a high tolerance carbon part*



*Wax plugs*



*Fiberglass mold*

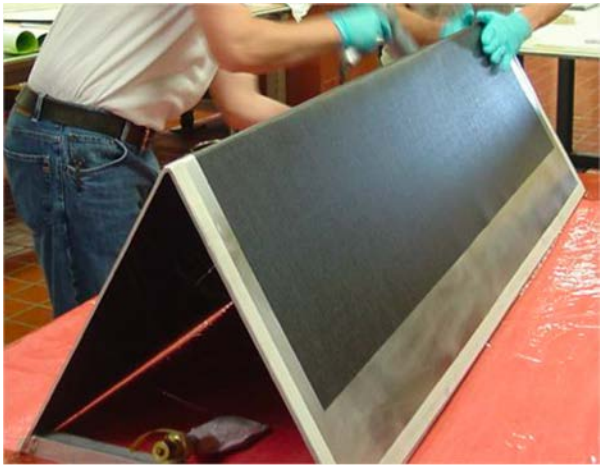


*Carbon epoxy propeller*

# Wing construction methods

- **Pre-preg molded construction** uses carbon cloth which is pre-impregnated with epoxy and laid into a mold which holds the structural shape under applied pressure
  - The pre-preg material must be refrigerated to prevent premature curing
  - Once it is exposed to room temperature or elevated temperature, it cures to a hard finish
  - The Boeing 787 has carbon epoxy wings that are cured in an autoclave

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*Pre-preg material is applied to a mold*



<https://brebeckcomposite.wordpress.com/tag/brebeck-composites-s-r-o-team/page/3/>

*The part is placed inside an autoclave (a high temperature pressure vessel) to cure*



*Finished parts are held to a much higher tolerance*